Ms. Ref. No.: **KNOSYS-D-26-21490R1**

Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

Journal: *Knowledge-Based Systems* (Elsevier)

To: Dr. Hang Yu, Senior Editor

Re: Response to Reviewers — Production Submission

Date: October 15, 2026

Dear Dr. Yu and Reviewers,

We thank you for the positive decision and for the careful reading of the revised manuscript. This letter accompanies the production files (LaTeX source and figures) and responds point by point to the three questions raised by Reviewer 1 in the acceptance decision. Reviewer 2 expressed no further concerns; we are grateful for their evaluation.

One textual change has been made to the manuscript: two sentences have been added to Section 2.4 (Parameter-efficient fine-tuning and forgetting) to address Reviewer 1's request for a comparison with sequential-task-adaptation methods. No other changes were made to the scientific content.

Respectfully submitted on behalf of all authors,

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Corresponding Author

Reviewer 2

Reviewer 2 approved the manuscript without further concerns. We are grateful for the thorough and constructive engagement throughout both rounds of review.

Reviewer 1

Reviewer 1 — Question 1:

> *Can the authors report a targeted, blinded manual check of short and verbose answers against the substring scorer, using the original fixed K_0 items, and explain whether any disagreements affect the key interpretations?*

The evaluation uses a bidirectional substring scorer: a generation is marked correct if either (a) the reference answer is a substring of the generated response, or (b) the generated response is a substring of the reference answer. Direction (a) handles verbose generated responses that contain the correct answer within a longer reply; direction (b) handles concise generated responses that partially match a longer reference alias. The K_0 item set is fixed at generation 0 using this same scorer, so the denominator is internally consistent with the numerator throughout all generations.

During the revision process, the scoring pipeline was subjected to a full audit that identified and corrected two implementation inconsistencies (described in Section 3.3 of the revised manuscript).

After those corrections, we conducted a targeted manual review of a stratified sample of K_0 items, covering both short reference answers (fewer than five tokens) and verbose reference answers (more than fifteen tokens). For the short-answer stratum, the bidirectional check correctly handled all sampled cases: short correct answers were captured by direction (b) when the generated response was also short, and by direction (a) when the generated response was verbose. For the verbose-answer stratum, the check correctly retained items where the full reference phrase appeared as a substring of the generation. No disagreements were found in the sampled items that would change the sign or statistical significance of any reported result. We therefore conclude that the key interpretations are not affected by answer-length variation.

Revised manuscript: Section 3.3 (Evaluation protocol), revised manuscript.

Reviewer 1 — Question 2:

> *Can the authors provide an immutable, internally consistent mapping from each reported condition to its actual run, seed, precision, K_0 item identifiers and generation-level counts, and regenerate the inferential statistics from those records?*

The complete experimental record is publicly archived at:

`https://doi.org/10.5281/zenodo.23145361`

The archive contains, for every reported condition, run directory, seed, floating-point precision (NF4 or bf16), the K_0 item set constructed at generation 0, and generation-level retention counts. The mapping from manuscript condition labels to archive paths is provided in the `README.md` at the repository root.

All inferential statistics reported in the manuscript (interaction estimates, confidence intervals, sign tests, and TOST equivalence tests) were regenerated directly from the per-seed `results_all.json` files archived at that DOI before this submission. The regeneration scripts are included in the archive under `scripts/audits_diagnostics/`.

Revised manuscript: Data Availability Statement and supplementary material, revised manuscript.

Reviewer 1 — Question 3:

> *The related-work discussion could further distinguish forgetting under sequential task adaptation from degradation of a recursively replaced data stream. A brief comparison with "Adaptive tree-like neural network: Overcoming catastrophic forgetting to classify streaming data with concept drifts" (Knowledge-Based Systems, 2024), focused on that distinction rather than a direct LLM baseline, would help clarify the scope of the contribution.*

We have added two sentences to Section 2.4 (Parameter-efficient fine-tuning and forgetting), immediately before the discussion of Biderman et al. [1]. The added passage reads:

> "Methods designed to overcome catastrophic forgetting under sequential task adaptation, such as the adaptive tree-like neural network of Wen et al. [2] which grows architecture branches for newly arriving concepts, address a different failure mode from the one studied here: rather than cross-task interference at discrete concept boundaries, the degradation we observe arises from a recursively replaced data stream in which each generation's synthetic outputs alter the distribution seen by the next."

Revised manuscript: Section 2.4, paragraph beginning "While classical continual learning. . . ", revised manuscript.

Production files. This submission includes (i) LaTeX source files (manuscript, bibliography, shared number definitions, and table bodies), (ii) figure files in PDF and PNG format at 300 DPI, and (iii) updated Highlights. The Highlights have been corrected to align with the manuscript: the bullet previously reading "arrests progressive loss" has been replaced by "A 50% synthetic exposure cut improved retention on both backbones tested," which accurately reflects the distinct patterns observed on Qwen and Gemma 3.

References

[1] Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, Mansheej Paul, Philip Greengard, Connor Jennings, Daniel King, Sam Havens, Vitaliy Chiley, Jonathan Frankle, Cody Blakeney, and John P. Cunningham. LoRA learns less and forgets less. *Transactions on Machine Learning Research (TMLR)*, 2024. Featured Certification. arXiv:2405.09673.

[2] YiMin Wen, Xiang Liu, and Hang Yu. Adaptive tree-like neural network: Overcoming catastrophic forgetting to classify streaming data with concept drifts. *Knowledge-Based Systems*, 293:111636, 2024. doi: 10.1016/j.knosys.2024.111636.